

THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

TEAM AWARD HOMEWORK SET #7

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2017
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Due Date: Wednesday Nov-22, 2017 in class

Number of points.....100
Your score.....

Award: Gift card to each member of the winning team

To be considered for the award, the team homework final report must be typed. Homework results should be presented in a logical and concise manner, with all steps in derivation processes [if any] shown very clearly. The ASPEN PFD with critical data must be printed in a full page and included with the homework solution report. Images of the Block Results as well as the Stream Results must also be clearly shown in the homework solution report. You may also want to include anything else (like external references) you believe will increase your chances of winning the award.

Remember: All members of the team that wins a total of three(3) awards will be given **100** points towards the final examination and will be excused from taking it.

HOMEWORK 7

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Contents

Part One: Problem ONE	3
I. Introduction.....	3
II. Physical Properties.....	3
Table 1-B Property Methods.....	4
III. Stream Inputs	4
Table 1-C Stream Inputs for FEEDGAS	4
Table 1-D Stream Inputs for LEANSOLV	4
IV. Problem 1 Flowsheet	5
Figure 1 Simulation of Amine-Guard Process.....	5
Figure 2 Absorber Block and Streams.....	6
Figure 3 Stripper Block and Streams.....	6
Figure 4 Heat Exchanger and Streams.....	7
Figure 5 Make-Up Streams.....	7
V. Block Inputs.....	7
Table 1-E Absorber.....	7
Table 1-F Stripper.....	8
Table 1-G Heat Exchanger	9
Table 1-H Pumps	10
Table 1-I Coolers	11
Table 1-J Flash Drums.....	11
VI. Stream Results	12
Table 1-K Final Results for Absorber.....	12
Table 1-L Final Results for Stripper.....	13
Table 1-M Final Results for Convergence.....	14
VII. Block Results	15
Absorber Results	15
Heat Exchanger Results	16
Reboiler Results	16
PUMP 2 Results	17
PUMP 3 Results	17
COOLER 1 Results	18
COOLER 2 Results	18
VIII. Operating Costs	19

	Table 1-H Operating Cost for Process	19
IX.	Conclusion	20
Works Cited		20
Appendices		21
X.	A-1 All Stream Results.....	21
XI.	A-2 All Stream Results continued	22

Part One: Problem ONE

Introduction

Commercial production of natural gas contains significant amounts of carbon dioxide, or acid gas, that threatens the overall economics of the process and the purity of the natural gas. However, the removal of CO₂ and other contaminants is crucial for the overall optimization of Liquefied Natural Gas (LNG) facilities. By sweetening the natural gas, the concentration of CO₂ is significantly reduced. Removal of CO₂ is essential in preventing corrosion of pipelines and gas processing equipment, as well as increase the heating value of the gas.

In the process, the feed gas flows into stage 15 of the absorber. Lean solution containing the amine solvent, MDEA, in water is pumped into the absorber at stage 1. As the solvent flows down the absorption column, it is in contact with the feed gas as the gas rises through the liquid. Once the gas reaches the top of the absorption column, most of the CO₂ has been removed from the feed gas leaving behind only 1% CO₂ in the now sweet gas. Next, the rich solvent exits at the bottom of the absorber where the stream is subsequently flashed to remove light hydrocarbons which may then be used as a fuel source elsewhere in the facility. The rich solvent stream flows through a heat exchanger where proper heat exchange between the lean solvent stream (at high Temperature) and the rich solvent stream (at lower Temperature) occurs and thus increasing the temperature of the rich solvent stream prior to entering the stripping column. The stripping column allows for the regeneration of the solvent. Any acid gas and fuel that was not removed by the flash drum will be removed at the top of the stripper column along with trace amount of solvent and large quantities of water. The vapor passes through a cooler to condense any solvent remaining in the acid gas stream. The acid gas is separated from the condensed liquid within another reflux drum. The separated liquid is pumped back to the top as a reflux. At the bottom of the stripping column the regenerated solution from the reboiler is pumped, cooled, and then recycled back to the absorption column as lean solvent.

Physical Properties

Table 1-A Components

Component ID	Type	Component Name	Alias
CO2	Conventional	CARBON-DIOXIDE	CO2
H2	Conventional	HYDROGEN	H2
H2O	Conventional	WATER	H2O
N2	Conventional	NITROGEN	N2
AR	Conventional	ARGON	AR
CH4	Conventional	METHANE	CH4
MDEA	Conventional	METHYL-DIETHANOLAMINE	C5H13NO2
MDEA+	Conventional	MDEA+	C5H14NO2+
H3O+	Conventional	H3O+	H3O+
OH-	Conventional	OH-	OH-

HCO3-	Conventional	HCO3-	HCO3-
CO3--	Conventional	CO3--	CO3-2

Table 1-B Property Methods

Property Methods & Options	
Method Filter	COMMON
Base Method	ELECTNRTL
Base Name	ELECTNRTL
Henry Components	GLOBAL
Chemistry ID	GLOBAL

Stream Inputs

Table 1-C Stream Inputs for FEEDGAS

Specification	Value
Temperature (°C)	40
Pressure (bar)	55
Vapor fraction	0
Flow Rate (kmol/hr)	2000
CO2	29.1
H2	10
H2O	59.8
N2	0.2
AR	0.3
CH4	0.1
MDEA	0.5
MDEA+	0
H3O+	0
OH-	0
HCO3-	0
CO3--	0

Table 1-D Stream Inputs for LEANSOLV

Specification	Value
Temperature (°C)	40
Pressure (bar)	55
Vapor fraction	0
Flow Rate (kmol/hr)	500
CO2	0
H2	0.0014
H2O	0
N2	0.8675
AR	0
CH4	0
MDEA	0

Figure 2 Absorber Block and Streams

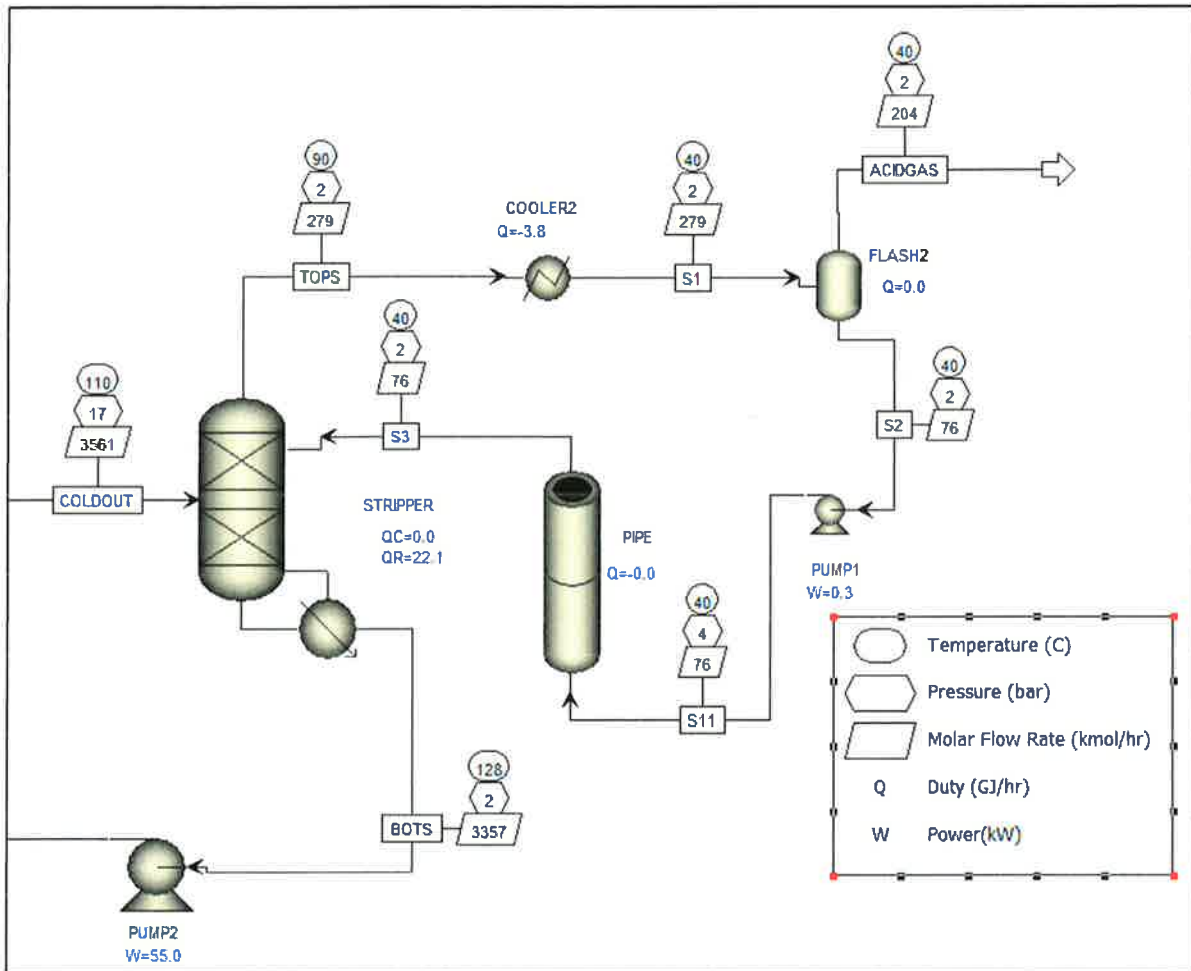


Figure 3 Stripper Block and Streams

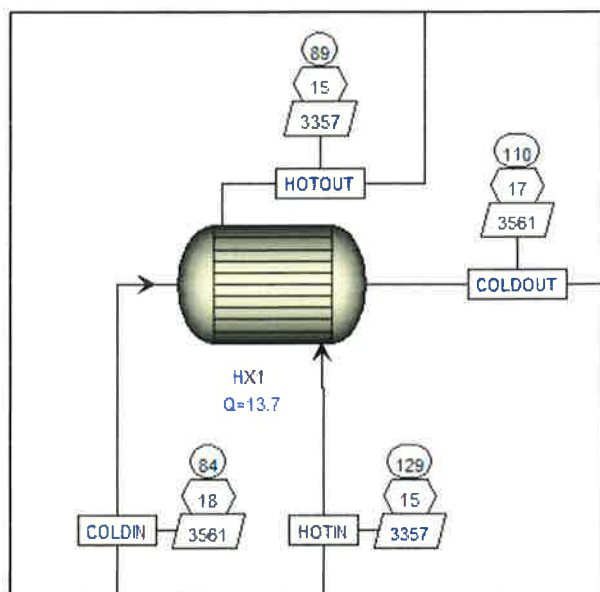


Figure 4 Heat Exchanger and Streams

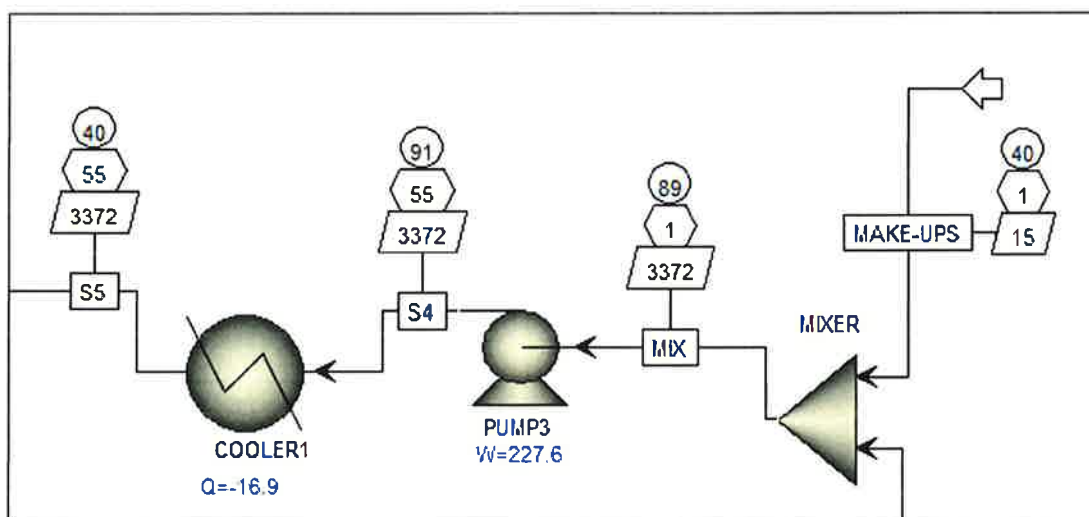


Figure 5 Make-Up Streams

Block Inputs

The objective of the absorber was to remove acid gas from the feed gas. The absorption column will operate at the feed gas pressure.

Table 1-E Absorber

Name	ABS
Property method	ELECNRTL
Henry's component list ID	GLOBAL

Electrolyte chemistry ID	GLOBAL
Free-water phase properties method	STEAM-TA
Water solubility method	3
Number of stages	15
Condenser	NONE
Reboiler	NONE
Number of phases	2
Free-water	NO
Top stage pressure [bar]	53
FEED STREAMS	
FEEDGAS	Above-Stage 16
LEANSOLV	On-Stage 1
PRODUCT STREAMS	
SWEETGAS	1 Vapor
RICHSOLV	15 Liquid
Pressure	
Top Stage/Condenser [bar]	53

Table 1-F Stripper

Name	STRIPPER
Property method	ELECNRTL
Henry's component list ID	GLOBAL
Electrolyte chemistry ID	GLOBAL
Use true species approach for electrolytes	NO
Free-water phase properties method	STEAM-TA
Water solubility method	3
Number of stages	18
Condenser	NONE
Reboiler	KETTLE
Number of phases	2
Free-water	NO
Top stage pressure [bar]	2
FEED STREAMS	
S3	On-stage 1
COLDOUT	Above-Stage 1
PRODUCT STREAMS	
TOPS	1 Vapor
BOTS	18 Liquid
Pressure	
Top Stage/Condenser [bar]	2
Pressure Drop for rest of column [bar]	0.3

The purpose of the stripper is to regenerate the lean solvent by stripping the acid gas in the rich solvent stream. The reboiler at the bottom of the stripping column heats up the rich solvent to produce the energy to reverse the chemical reaction to free the acid gas from the amine solution and to generate steam to strip acid gas from the solution.

Table 1-G Heat Exchanger

Name	HX1
Hot side property method	ELECNRTL
Hot side Henry's component list ID	GLOBAL
Hot side electrolyte chemistry ID	GLOBAL
Hot side use true species approach for electrolytes	NO
Hot side free-water phase properties method	STEAM-TA
Hot side water solubility method	3
Cold side property method	ELECNRTL
Cold side Henry's component list ID	GLOBAL
Cold side electrolyte chemistry ID	GLOBAL
Cold side use true species approach for electrolytes	NO
Cold side free-water phase properties method	STEAM-TA
Cold side water solubility method	3
Exchanger specification type	COLD OUTLET TEMP
Exchanger specification	110
Units of exchanger specification	C
Minimum temperature approach [C]	10
Hot side outlet pressure [bar]	-0.344737865
Cold side outlet pressure [bar]	-0.689475729

The amine heat exchanger allows for the preheating of the rich solvent to the regenerator. The rich solution flowed through the **tube side** and the lean solution flowed through the **shell side**. The hot stream (lean solvent stream) flows through the shell side since the fluid is hotter and can be cooled by surrounding air as well as the colder fluid flowing through the tube side. The lean solvent inlet temperature was around 110 °C and the rich solvent outlet temperature was around 90 °C.

Table 1-H Pumps

Name	PUMP1	PUMP2	PUMP3
Property method	ELECNRTL	ELECNRTL	ELECNRTL
Henry's component list ID	GLOBAL	GLOBAL	GLOBAL
Electrolyte chemistry ID	GLOBAL	GLOBAL	GLOBAL
Use true species approach for electrolytes	NO	NO	NO
Free-water phase properties method	STEAM-TA	STEAM-TA	STEAM-TA
Water solubility method	3	3	3
Specified discharge pressure [bar]	4	15	55

To estimate the discharge pressure of the reflux pump, the assumption that the reflux pump positioned at zero elevation was made. Assuming that the space between each stage was 1 m, the height of the column was calculated to be approximately 18 m. Since most of the solution is composed of water, the density of water was taken into consideration. Then, to calculate the head loss pressure, the following calculations were made:

$$D \cdot V \cdot g$$

The above equation would give force. However, by dividing the equation by the area would produce the head loss pressure.

$$D \cdot V \cdot g / A$$

By substituting the V (volume) by A (area) multiplied by the h (height) would give:

$$D \cdot A \cdot h \cdot g / A$$

Finally, by canceling the A, pressure can be achieved by:

$$D \cdot h \cdot g$$

Where $D = 1000 \text{ kg/m}^3$

$$h = 18 \text{ m}$$

$$g = 10 \text{ m/s}^2$$

Plugging the above values, the head loss pressure is 180,000 Pa or approximately 2 bar. Since the pressure of the condenser was at 2 bar and the head loss pressure was about 2 bar, the discharge pressure of PUMP1 was approximated at 4 bar.

Table 1-I Coolers

Name	COOLER1	COOLER2
Property method	ELECNRTL	ELECNRTL
Henry's component list ID	GLOBAL	GLOBAL
Electrolyte chemistry ID	GLOBAL	GLOBAL
Use true species approach for electrolytes	NO	NO
Free-water phase properties method	STEAM-TA	STEAM-TA
Water solubility method	3	3
Specified pressure [bar]	0	0
Specified temperature [C]	40	40

Since absorber column favors solution at high pressure and low temperature, the lean solvent is cooled by COOLER 1 after it was pressurized to 55 bar.

Table 1-J Flash Drums

Name	FLASH1	FLASH2
Property method	ELECNRTL	ELECNRTL
Henry's component list ID	GLOBAL	GLOBAL
Electrolyte chemistry ID	GLOBAL	GLOBAL
Use true species approach for electrolytes	NO	NO
Free-water phase properties method	STEAM-TA	STEAM-TA
Water solubility method	3	3
Temperature [C]	84	
Pressure [bar]	18	2
Specified heat duty [GJ/hr]		0
Valid Phases	Vapor-Liquid	Vapor-Liquid

FLASH1 removes the dissolved gas and entrained fuel. FLASH2 recovers the water condensed in the condenser.

Stream Results

Table 1-K Final Results for Absorber

** SEE FIGURE 2

	FEEDGAS	LEANSOLV	SWEETGAS	RICH SOLV
Mole Frac				
CO	0.291	0	0.3139	0.0043
CO2	0.1	0.0014	0.0102	0.0523
H2	0.598	0	0.6620	0.0002
H2O	0.002	0.8675	0.0056	0.8185
N2	0.003	0	0.0033	0.0000
AR	0.001	0	0.0010	0.0000
CH4	0.005	0	0.0040	0.0008
MDEA	0	0.1311	0.00	0.1240
MDEA+	0	0	0.00	0.00
H3O+	0	0	0.00	0.00
OH-	0	0	0.00	0.00
HCO3-	0	0	0.00	0.00
CO3--	0	0	0.00	0.00
Total Flow kmol/hr	2000.00	3371.93	1805.55	3566.38
Total Flow kg/hr	27995.47	105583.00	19633.05	113945.00
Temperature C	40	40	69	84
Pressure bar	55	55	53.00	53
Vapor Frac	0.99	0	1.00	0
Liquid Frac	0.00	1	0.00	1

Table 1-L Final Results for Stripper

** SEE FIGURE 3

	COLDOUT	BOTS	TOPS
Mole Frac			
CO	0.0033	0.0000	0.0425
CO2	0.0520	0.0011	0.6495
H2	0.0000	0.0000	0.0002
H2O	0.8197	0.8672	0.2977
N2	0.0000	0.0000	0.0000
AR	0.0000	0.0000	0.0004
CH4	0.0007	0.0000	0.0096
MDEA	0.1241	0.1317	0.0001
MDEA+	0.00	0.00	0.00
H3O+	0.00	0.00	0.00
OH-	0.00	0.00	0.00
HCO3-	0.00	0.00	0.00
CO3--	0.00	0.00	0.00
Total Flow kmol/hr	3560.70	3356.97	279.36
Total Flow kg/hr	113787.00	105287.00	9866.87
Temperature C	110	128.3529	90
Pressure bar	17.31052	2.3	2
Vapor Frac	0.02127	0	1
Liquid Frac	1	1	0

Table 1-M Final Results for Convergence**** SEE FIGURE 2, 3, 5**

	LEANSOLV	S5	BOTS	SWEETGAS
Mole Frac				
CO	0	0.00	0.0000	0.3139
CO2	0.0014	0.0011	0.0011	0.0102
H2	0	0.00	0.0000	0.6620
H2O	0.8675	0.8678	0.8672	0.0056
N2	0	0.00	0.0000	0.0033
AR	0	0.00	0.0000	0.0010
CH4	0	0.00	0.0000	0.0040
MDEA	0.1311	0.1311	0.1317	0.00
MDEA+	0	0.00	0.00	0.00
H3O+	0	0.00	0.00	0.00
OH-	0	0.00	0.00	0.00
HCO3-	0	0.00	0.00	0.00
CO3--	0	0.00	0.00	0.00
Total Flow kmol/hr	3371.93	3371.94	3356.97	1805.55
Total Flow kg/hr	105583.00	105557.00	105287.00	19633.05
Temperature C	40	40	128.3529	69
Pressure bar	55	55	2.3	53.00
Vapor Frac	0	0	0	1.00
Liquid Frac	1	1	1	0.00

When converging the simulation, the biggest take away was to obtain a CO₂ molar fraction that met the requirements. As seen in **Table 1-M**, the final results show the convergence for the LEANSOLV stream and S5 stream to within a 2% difference. The stream S5 is the stream exiting the COOLER1 after the Make-Up Amine has been added. As seen above, in the BOTS stream leaving the absorber and S5, that make-up amine made up the difference to get the total molar flowrate to once again equal the LEANSOLV stream inlet in order to meet the 1% CO₂ requisite in the absorber

Block Results

Absorber Results

Condenser / Top stage performance

Name	Value	Units
Temperature	68.9113	C
Subcooled temperature		
Heat duty	0	GJ/hr
Subcooled duty		
Distillate rate	1805.55	kmol/hr
Reflux rate	3510.85	kmol/hr
Reflux ratio	1.94448	
Free water distillate rate		
► Free water reflux ratio		

Reboiler / Bottom stage performance

Name	Value	Units
Temperature	84.3617	C
Heat duty	0	GJ/hr
Bottoms rate	3566.38	kmol/hr
► Boilup rate	1991.82	kmol/hr
Boilup ratio	0.558498	
Bottoms to feed ratio		

Heat Exchanger Results

Heatx results

Calculation Model	Shortcut			
	Inlet		Outlet	
Hot stream:	HOTIN		HOTOUT	
Temperature:	128.794	C	88.9001	C
Pressure:	15	bar	14.6553	bar
Vapor fraction:	0		0	
1st liquid / Total liquid	1		1	
Cold stream:	COLDIN		COLDOUT	
Temperature:	84	C	110	C
Pressure:	18	bar	17.3105	bar
Vapor fraction:	0		0.02127	
1st liquid / Total liquid	1		1	
Heat duty:	13.6601	GJ/hr		

Reboiler Results

Reboiler / Bottom stage performance

Name	Value	Units
Temperature	128.353	C
Heat duty	22.1034	GJ/hr
Bottoms rate	3356.97	kmol/hr
Boilup rate	455.239	kmol/hr
Boilup ratio	0.13561	
Bottoms to feed ratio		

PUMP 2 Results

Fluid power:	39.0805505	kW	▼
Brake power:	54.9747	kW	▼
Electricity:	54.9747	kW	▼
Volumetric flow rate:	110.78	cum/hr	▼
Pressure change:	12.7	bar	▼
NPSH available:	0	J/kg	▼
NPSH required:			▼
Head developed:	1336.25	J/kg	▼
Pump efficiency used:	0.710882		
Net work required:	54.9747	kW	▼
Outlet pressure:	15	bar	▼

PUMP 3 Results

Fluid power:	161.145012	kW	▼
Brake power:	227.588	kW	▼
Electricity:	227.588	kW	▼
Volumetric flow rate:	107.43	cum/hr	▼
Pressure change:	54	bar	▼
NPSH available:	41.6591	J/kg	▼
NPSH required:			▼
Head developed:	5495.8	J/kg	▼
Pump efficiency used:	0.708056		
Net work required:	227.588	kW	▼
Outlet pressure:	55	bar	▼

COOLER 1 Results

Outlet temperature:	40	C	▼
Outlet pressure:	55	bar	▼
Vapor fraction:	0		
Heat duty:	-16.862	GJ/hr	▼
Net duty:	-16.862	GJ/hr	▼
1st liquid / Total liquid:	1		
Pressure-drop correlation parameter:			

COOLER 2 Results

Outlet temperature:	40	C	▼
Outlet pressure:	2	bar	▼
Vapor fraction:	0.729267		
Heat duty:	-3.78538	GJ/hr	▼
Net duty:	-3.78538	GJ/hr	▼
1st liquid / Total liquid:	1		
Pressure-drop correlation parameter:			

Operating Costs

**Cost Estimates were conducted by considering one year of operation with the process being fully operational 90% of the time or 8,000 Hours on Stream (HOS).*

Table 1-H Operating Cost for Process

UTILITIES				
Unit	Duty (GJ)	HOS	Utility Cost (\$/GJ)	Annual Utility Cost
FLASH1	0.0929	8000	\$ 5.5	\$ 4,087.60
FLASH2	0.1		\$ 5.5	\$ 4,400.00
COOLER1	16.9		\$ 0.2	\$ 27,040.00
COOLER2	3.8		\$ 0.2	\$ 6,080.00
REBOILER	22.1		\$ 5.5	\$ 972,400.00
ELECTRICITY				
Unit	Work (kW)	HOS	Electrical Cost (\$/kWh)	Annual Electrical Cost
PUMP1	0.3	8000	\$ 0.06	\$ 144.00
PUMP2	55		\$ 0.06	\$ 26,400.00
PUMP3	227.6		\$ 0.06	\$ 109,248.00
TOTAL				\$ 1,149,799.60

Conclusion

In order to verify the feasibility of the proposed natural gas processing facility, an economic analysis was conducted taking into consideration only the annualized operational expenses. The proposed project requires a total of three pumps, two flash drums, two coolers, a shell and tube heat exchanger, and a kettle reboiler. Although each process equipment plays a crucial role in the integrity of the process, the utility requirements per equipment vary in magnitude. From **Table 1-H** it can be seen that every piece of equipment pertains to different utility requirements, for instance, COOLER1 requires \$27,040.00/year which accounts for a mere 2.4% of the total annualized operational expenses, whereas, the REBIOLER and PUMP3 require \$972,400.00 and \$109,248.00 amounting to 84.6% and 9.50% of the total operational expenses respectively. Evidently, proper sizing of the reboiler is essential to minimize operational costs, with this in mind, it is the recommendation of this group to increase the rich solvent stream temperature to 110C prior to entering the stripping column in order to optimize the duty requirement from the reboiler along with maintaining a 6.615 H₂O to lean solvent ratio and a 0.01 CO₂ load within the recycled lean solvent stream in order to achieve the 1% CO₂ in sweet gas specification requirement.

Works Cited

[1] Dr. Caracotsios, CHE433 Process Simulation with Aspen Plus, Team Award Homework Set #7

Appendices

A-1 All Stream Results

	ACIDGAS	BOTS	COLDIN	COLDOUT	FEEDGAS	FUEL	HOTIN	HOTOUT	LEANSOLV	MAKE-UPS	MIX	RICH SOLV
Mole Frac												
CO	0.0583	0.0000	0.0033	0.0033	0.2910	0.5985	0.0000	0.0000	0.0000	0.0000	0.0000	0.0043
CO2	0.8902	0.0011	0.0520	0.0520	0.1000	0.2275	0.0011	0.0011	0.0014	0.0000	0.0011	0.0523
H2	0.0003	0.0000	0.0000	0.0000	0.5980	0.1304	0.0000	0.0000	0.0000	0.0000	0.0000	0.0002
H2O	0.0377	0.8672	0.8197	0.8197	0.0020	0.0282	0.8672	0.8672	0.8675	0.9999	0.8678	0.8185
N2	0.0000	0.0000	0.0000	0.0000	0.0030	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
AR	0.0006	0.0000	0.0000	0.0000	0.0010	0.0024	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CH4	0.0130	0.0000	0.0007	0.0007	0.0050	0.0126	0.0000	0.0000	0.0000	0.0000	0.0000	0.0008
MDEA	0.0000	0.1317	0.1241	0.1241	0.0000	0.0000	0.1317	0.1317	0.1311	0.0001	0.1311	0.1240
MDEA+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H3O+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
OH-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HCO3-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CO3--	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Flow kmol/hr	203.73	3356.97	3560.70	3560.70	2000.00	5.68	3356.97	3356.97	3371.93	14.97	3371.94	3566.18
Total Flow kg/hr	8499.49	105287.00	113787.00	113787.00	27995.47	158.21	105287.00	105287.00	105583.00	269.94	105557.00	113945.00
Temperature C	40	128	84	110	40	84	129	89	40	40	89	84
Pressure bar	2	2.3	18	17.31052	55	18	15	14.65526	55	1	1	53
Vapor Frac	1	0	0	0.02127	0.999856	1	0	0	0	0	0	0
Liquid Frac	0	1	1	1	0	0	1	1	1	1	1	1

A-2 All Stream Results continued

	S1	S2	S3	S4	S5	S11	SWEETGAS	TOPS
Mole Frac								
CO	0.0425	0.0002	0.0002	0.0000	0.0000	0.0002	0.3139	0.0425
CO2	0.6495	0.0011	0.0011	0.0011	0.0011	0.0011	0.0102	0.6495
H2	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	0.6620	0.0002
H2O	0.2977	0.9980	0.9980	0.8678	0.8678	0.9980	0.0056	0.2977
N2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0033	0.0000
AR	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0010	0.0004
CH4	0.0096	0.0004	0.0004	0.0000	0.0000	0.0004	0.0040	0.0096
MDEA	0.0001	0.0003	0.0003	0.1311	0.1311	0.0003	0.0000	0.0001
MDEA+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
H3O+	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
OH-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HCO3-	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CO3--	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Flow kmol/hr	279.36	75.63	75.63	3371.94	3371.94	75.63	1805.55	279.36
Total Flow kg/hr	9866.87	1367.38	1367.29	105557.00	105557.00	1367.38	19633.05	9866.87
Temperature C	40	40	40	91	40	40	69	90
Pressure bar	2	2	2.253403	55	55	4	53	2
Vapor Frac	0.7292673	0	0	0	0	0	1	1
Liquid Frac	0.2707327	1	1	1	1	1	0	0